EVALUATION

I Multiple Choice Questions

1. The wavelength λ_e of an electron and λ_p of a photon of same energy E are related by (NEET 2013)

a) $\lambda_p \propto \lambda_e$ b) $\lambda_p \propto \sqrt{\lambda_e}$

c) $\lambda_p \propto \frac{1}{\sqrt{\lambda_e}}$ d) $\lambda_p \propto \lambda_e^2$

2. In an electron microscope, the electrons are accelerated by a voltage of 14 kV. If the voltage is changed to 224 kV, then the de Broglie wavelength associated with the electrons would

a) increase by 2 times

b) decrease by 2 times

c) decrease by 4 times

d) increase by 4 times

3. The wave associated with a moving particle of mass 3×10^{-6} g has the same wavelength as an electron moving with a velocity $6\times10^{6}\,\text{m s}^{-1}$. The velocity of the particle is

a) $1.82\times10^{-18}\,\text{m s}^{-1}$

b) $9\times10^{-2}\,\text{m s}^{-1}$

c) $3\times10^{-31}\,\text{m s}^{-1}$

d) $1.82\times10^{-15}\,\text{m s}^{-1}$

4. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $\frac{V}{4}$. The threshold wavelength for the metallic surface is (NEET 2016)

a) 4λ b) 5λ

c) $\frac{5}{2}\lambda$ d) 3λ

5. If a light of wavelength 330 nm is incident on a metal with work function 3.55 eV, the electrons are emitted. Then the wavelength of the wave associated with the emitted electron is (Take $h = 6.6 \times 10^{-34}$ Js)

a) $< 2.75\times10^{-9}\,\text{m}$ b) $\geq 2.75\times10^{-9}\,\text{m}$

c) $\leq 2.75\times10^{-12}\,\text{m}$ d) $< 2.5\times10^{-10}\,\text{m}$

6. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and $\lambda/2$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the material is (NEET 2015)

a) $\frac{hc}{\lambda}$ b) $\frac{2hc}{\lambda}$

c) $\frac{hc}{3\lambda}$ d) $\frac{hc}{2\lambda}$

7. In photoelectric emission, a radiation whose frequency is 4 times threshold frequency of a certain metal is incident on the metal. Then the maximum possible velocity of the emitted electron will be

a) $\sqrt{\frac{hv_0}{m}}$ b) $\sqrt{\frac{6hv_0}{m}}$

c) $2\sqrt{\frac{hv_0}{m}}$ d) $\sqrt{\frac{hv_0}{2m}}$

8. Two radiations with photon energies 0.9 eV and 3.3 eV respectively are falling on a metallic surface successively. If the work function of the metal is 0.6 eV, then the ratio of maximum speeds of emitted electrons in the two cases will be

a) 1:4 b) 1:3

c) 1:1 d)1:9

XII_Physics_Vol-2_UNIT-8_EM.indd 135 03-03-2022 15:21:06